

Problem-Solution Fit

Date	15 JUNE 2026
Team ID	XXXXXX
Project Name	Sustainable Waste Management Assistant Using Generative AI
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Urban residents, college students, households in smart cities People motivated to recycle but unsure of correct disposal methods	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited budget (prefer free solutions) • Device constraints (must work on mobile and desktop) • Limited technical knowledge (needs simple UI)	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • Government websites: Pros – official info; Cons – complex, hard to navigate • Manual recycling guides: Pros – detailed; Cons – static, not personalized • Paid apps: Pros – advanced features; Cons – cost barrier
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2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Confusion about whether waste is recyclable or hazardous (daily) • Difficulty locating nearby collection centers (weekly) • Lack of awareness about eco-friendly alternatives (frequent)	9. PROBLEM ROOT / CAUSE [RC] • Lack of accessible, real-time guidance on waste classification • Fragmented information across multiple sources • Low civic awareness about sustainable disposal practices	7. BEHAVIOR + ITS INTENSITY [BE] • Scans waste items occasionally (moderate intensity) • Checks maps when unsure (low to moderate) • Reviews dashboard to track habits (moderate to high for eco-conscious users)
3. TRIGGERS TO ACT [TR] • Encountering a waste item (battery, plastic bottle, e-waste) • Desire to avoid environmental harm • Civic awareness campaigns or sustainability drives	10. YOUR SOLUTION [SL] • AI-powered web app that classifies waste instantly • Provides disposal instructions, hazard warnings, recycling steps, and eco-friendly suggestions • Maps nearby collection centers with details • Tracks user habits with analytics dashboards	8. CHANNELS OF BEHAVIOR [CH] ONLINE • Online: Web app, mobile browser, social media awareness • Offline: Visiting collection centers, community recycling drives
4. EMOTIONS BEFORE / AFTER [EM] • <i>Before:</i> Confused, anxious, guilty about improper disposal • <i>After:</i> Relieved, confident, proud of eco-friendly choices		OFFLINE